

Semantics

"Don't argue over semantics." ??

What does it mean? शब्दों का अर्थ

• Study of the meaning of words, phrases, & sentences.

• Semantic analysis

⇓

conventional meaning of linguistic forms, not necessarily what an individual thinks or intends

that is a shared meaning across all speakers of a language competent

Different types of meaning

Referential meaning

• Basic, essential components of meaning

• literal use of the word

e.g. श्वेत means the animal that has four limbs and a tail and is a primate, likes banana, associated with Hanuman etc.

Associative meaning

• The word श्वेत can mean different things to different people.

• It is associated to persons who destroy other's property, or to someone who doesn't do one's job in time, or may be associated with Hanuman, etc.

• Such meanings are not referential, but associative.

Semantic feature

actress: [+female] [+performer]

- Basic element of a word ^{by} [+animate] [+domesticated] [-wild] [-feline]
- More precisely: component of a concept that is associated with any grammatical unit like word

example: ① cat

[+animate] [+domesticated] [feline]

② leopard

[+animate] [-domesticated] [feline]

③ actor

[+male] [+performer]

Can you expand the list of features?

- To understand word meaning
- Mental representation of words - tagging
- NLP & AI

- compare meaning mathematically
- cluster related concepts
- model semantics in AI / LLM

Vector space of semantic feature

feature	man	woman	♂ dog	♀ cat
[+human]	1	1	0	0
[+female]	0	1	0	1
[+animal]	0	0	1	1

man: [1,0,0] woman: [1,1,0] dog: [0,0,1] cat: [0,1,1] ← 3-D space

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Lexical relations

- The relationships of meaning between words
- Words are described not in terms of their component features, but in terms of the relationship with other words
 - Synonymy • Homonymy • Homophony • Antonymy • Polysemy

* **Synonymy**: synonyms, e.g. big / large / huge, funny / comical
• often can be replaced one for the other, but not always

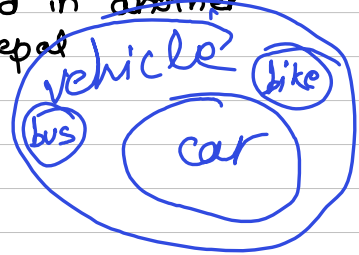
* **Antonymy**: antonyms, e.g. girl / boy, live / die, high / low
• one means the opposite of the other

- Gradable: tall / short, sad / happy
- Non-gradable: yes / no, alive / dead } aka complementary pairs
- Reversives: dress / undress, rise / fall, pack / unpack, enter / exit
↳ "do the opposite of"

* **Hyponymy**: meaning of one form included in another
vehicle / car, animal / tiger, country / Nepal

↳ car is a hyponym of vehicle

- Co-hyponyms: car, bus, bike
- Superordinate: vehicle, animal, country



* **Prototype**: "a characteristic instance"

↳ helps explain the meaning of certain word(s) / category

e.g. sparrow is a prototype of bird, but ostrich is not
t-shirt is a prototype of clothing, but shoes is not

* Homophones: different forms, same pronunciation, most likely different meanings
e.g. to/too/two, pale/pail, meat/meet,
right/write, flour/flower

* Homonyms: same form, same pronunciation, different meaning
e.g. bank (financial institution)
bank (riverside)

mole (on skin) - mole (small animal)

race (contest of speed) - race (ethnic group)

bat (flying creature) - bat (used in sports)

• Homonyms have separate histories and meanings, no connection.

• Separate dictionary entries

* Polysemy: same form, related meanings
e.g. head

• a part on top of the body

• froth on top of a beer glass

• person at the top of a company

• Single dictionary entry with multiple meanings listed

* Metonymy: a figure of speech

an object/concept is referred to by something that is closely related — content-container relation, whole-part relation, or a representative-symbol relation.

e.g. bottle/water, house/roof, the PM / ^{the} Singhadurbar
kettle/water car / tank

* Metaphor: a figure of speech in which a term is transferred from the object it ordinarily designates to an object it may designate only by implicit comparison or analogy
e.g. ^{दोपहर} ^{उज्ज} ^{हुन} / dawn of her career

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Sentence level

* Entailment

p: Cat killed rat.

q: Rat died.

	p	q
i	T	T
ii	F	T or F
iii	F	F
iv	T or F	T

i) When p is true, q is true.

ii) When p is false, q can be true or false.

iii) When q is false, p is false.

iv) When q is true, p can be true or false.

a square $\xrightarrow{\text{entails}}$ a rectangle

not square $\nrightarrow$ not rectangle

* Equivalence

A and B are in equivalence relⁿ if they entail each other

e.g. [John fucked Mary.]
[Mary fucked John.]

[Ram owns this book.]
[This book belongs to Ram.]

* Contradiction

A and B are in contradiction if each entails that the other is false, i.e. A entails the negation of B or vice versa.

e.g. No one is happy.

Someone is happy.

* Presupposition

- embedded assumption is the sentence
- The presupposition remains despite negation

The moon in my pocket is glowing hot.

The moon in my pocket is not glowing hot.

↳ "there is a moon in my pocket"

↳ may or may not be true → we are making supposition

Is the moon in my pocket glowing hot?

P: The emperor of Mars is my friend.

Q: There is an emperor of Mars.

	P	Q
i	T	→ T
ii	F	→ T
iii	T or F	→ T

* Presupposition failure

"Have you stopped bunking classes on Mondays?"

- no good answer to it

* Implicature → we'll see in Pragmatics

- not an entailment but inference you draw

eg. "Can you pass me the pen?"

implicature: The speaker wants you to pass her the pen.

Teaser for the next class:

- I'm glad I'm a man, and so is Lola.
- I saw her duck.
- Everyone in this class speaks at least two languages.
- I shot an elephant in my pyjamas.

Ambiguity

- if you can infer more than one plausible meaning
- can arise at different levels from phonetics to ^{morphology,} semantics, syntax

① Lexical ambiguity ← Semantic ambiguity

- when a word (or morpheme) is ambiguous

* polysemy * homonym * homophony

e.g. He likes it because of its bark.

No dear! I can't get you a bad!

I saw her duck. He kicked the bucket.

② Scope ambiguity ← Syntactic ambiguity... ^{more} later

Every teacher has a favorite student.

inverse scope: One student who is liked by all teachers.

surface scope: All teachers have at least one student they like, not necessarily the same student.

From previous class: What's the gender of a table?

	the sun:	el sol ^M	:	die Sonne ^F
N	the moon:	la luna ^F	:	der Mond ^M
	the table:	la mesa ^F	:	die Tabelle ^F der Tisch ^M
M	the boy:	el niño ^M	:	der Junge ^M
	the girl:	la niña ^F	:	das Mädchen ^F
F	the woman:	la mujer ^F	:	die Frau ^F

Syntax

- the rules that govern how words can be combined to form phrases and sentences

Karna constructed a chariot.

vs.

Constructed Karna chariot a.

- The rules to form a sentence refer to categories or types of words (e.g. nouns, verbs), not specific words themselves (e.g. Karna, created).

For example: "A sentence can be created from a noun and a verb." instead of "... from Karna and constructed"

- Generality of syntactic rules

- once speakers of a language know the category of a word, they can form a sentence using that word following the rule that ^{word}category follows

e.g. if told that "rostalizole" is a new medicine recently discovered, a new sentence can be formed as follows:

- If you have an upset stomach, take a capsule of rostalizole before meal.

- Syntactic categories (or parts of speech; pos)

Noun (N), Verb (V), Adjective (A), Determiner (D), and Preposition (P) $\Rightarrow$ major word categories

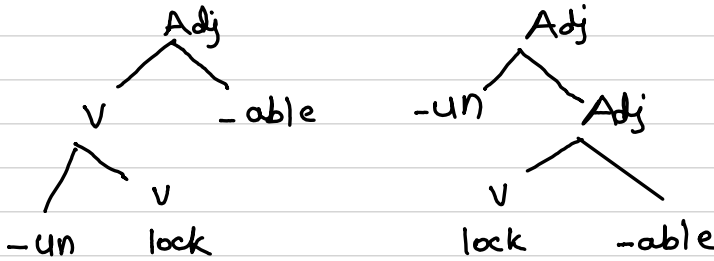
Other:

Degree words (Deg): too, so, very, much, quite

Auxiliary (Aux): will, could, must, be, have, do

Conjunction (Con): and, or, but, nonetheless, however

unlockable $\Rightarrow$ unlock + able
 $\Rightarrow$ un + lockable



i.e. "unlockable" is not ^{just} a series of 3 morphemes, it's how they are arranged and interpreted.

Similarly, in sentences, they're not just sequences of words, but how they are assembled and interpreted.

- Madan walked up the hill.
- Muna looked up the dictionary.

Meaningful — familiar — Let's go to the dance floor.
 — unfamiliar — Their cockroach is ^{dancing} on a hot air balloon.
 Impossible — Let's the dance floor go to.

- not about memorising ^{past sentences} or ^{their} meanings
 (see categories y'day)

It's not about meaning. Where does meaning come from?

- Colorless green ideas sleep furiously.
- Green ideas furiously sleep colorless.

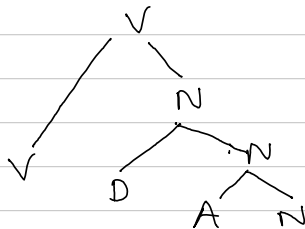
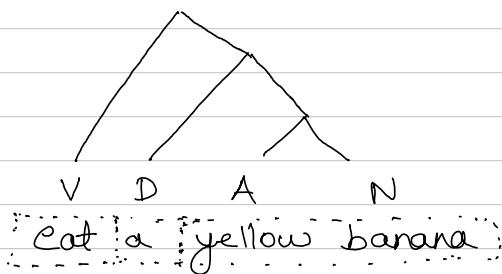
Competence vs Performance

- Let's the dance floor go to.

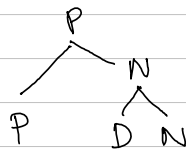
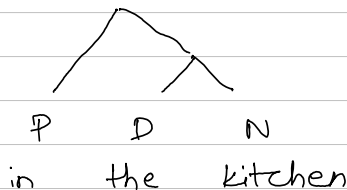
Normally people don't speak in full sentence, paragraph.
There are pauses and fillers. (except Noam Chomsky lol)
i.e. Performance

- Let's go to the dancefloor.
- Let's go to the dancefloor and spend the night dancing.
- She says, "Let's go to the"
- In my imagination, she says, "..."

This can go on and on to infinity
i.e. Competence



You should eat a yellow banana in the kitchen.



I'll walk up the stairs.

I'll wake up the kids.

Phrase structure & Phrase structure rules

PS: The way parts of sentences/phrases are organised and related to one another

- a concept closely related to Chomsky's Transformational Generative Grammar

PS rules: describe the syntax of a language, how the constituent parts of the phrases are organized

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syntactic categories, including lexical and phrasal categories

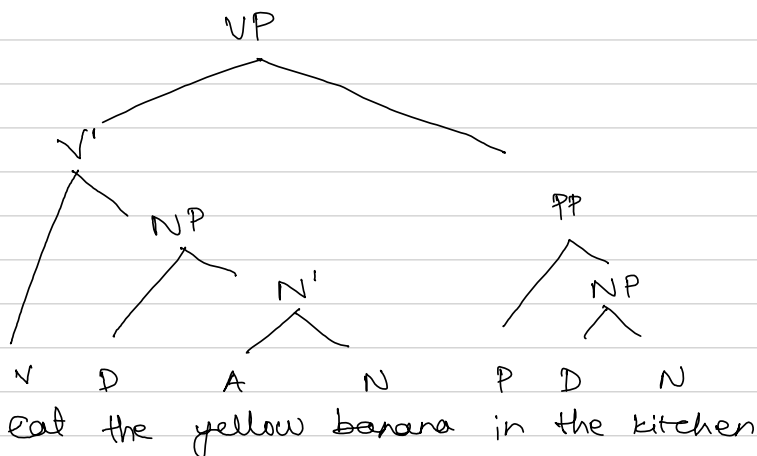
e.g. $S \rightarrow NP VP$

i.e. A sentence consists of a noun phrase and a verb phrase.

$NP \rightarrow (Det) N_1$

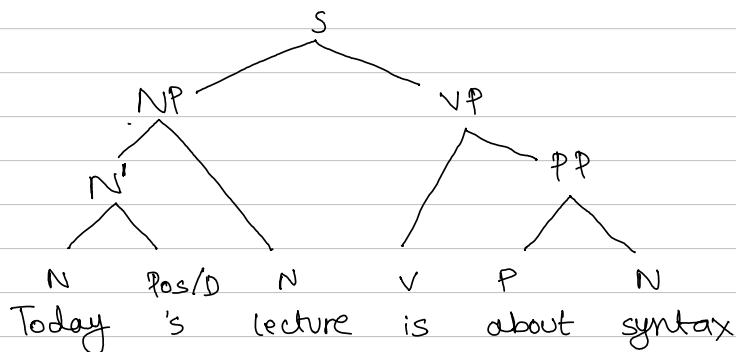
$N_1 \rightarrow (AP) N_1 (PP)$

where the constituents in round brackets are optional.



* highest node with a given label $\rightarrow$ Phrase (VP, NP, PP)

- In the example above, PP is modifying a VP.
- Below, VP modifies a NP:



- NP can contain PP; PP can contain NP.
(Lecture about syntax) (In the morning)

• Recursion

$\hookrightarrow$ because NP can contain PP which can contain NP, sentences can become infinitely long.



competence

But are they? $\Rightarrow$ performance

Topicalization

- A method of establishing an expression as the topic of a sentence (or clause) by moving it to the front of the sentence rather than in its standard position in the later part of the sentence.

- movement of determiners, prepositions, verbs

- whatever can be topicalized is considered "constituent"

e.g.: She gave him an apple → An apple she gave him.

They work for money. → For money, they work.

She works hard despite her illness. → Despite her illness she works hard.

Can I have that pie? → That pie, can I have?